

DIGITAL EMERGENCY TRIAGE SYSTEM

Pseudocode Design Document — Paragraph Method

Week 0 Submission | Emergency Medicine & AI Design

1. Overview

The Digital Emergency Triage System is an algorithm designed to support emergency department nurses in assigning patients to one of five acuity levels, consistent with the Canadian Triage and Acuity Scale (CTAS). The system does not replace clinical judgement; it functions as a structured decision aid that processes patient data at the point of arrival and suggests a risk category. The nurse retains full authority to accept, modify, or override any system-generated recommendation before it is recorded in the electronic patient record. The entire pipeline, from data collection through to escalation, is described in plain paragraphs across the five stages below.

2. Stage One — Collecting Patient Data

When a patient arrives at the emergency department, the first task of the system is to gather a complete and validated intake record. A bedside tablet or registration kiosk prompts the nurse or administrative clerk to enter the patient's age, biological sex, and the time of arrival. A chief complaint is entered as free text, describing in the patient's or nurse's own words the primary reason for attending. Alongside this, automated peripheral devices upload vital signs directly into the system wherever possible: these include the respiratory rate in breaths per minute, the heart rate in beats per minute, the systolic and diastolic blood pressure in mmHg, the peripheral oxygen saturation as a percentage, and the body temperature in degrees Celsius. Where automated capture is unavailable, the nurse enters values manually. The Glasgow Coma Scale score and AVPU consciousness level are assessed and entered by the nurse, as is the patient's self-reported pain score on a zero-to-ten scale. Before the record proceeds any further, the system checks that every mandatory field has been completed. If any field is missing, the nurse receives an alert and the record is held until the gap is resolved, preventing incomplete data from propagating through the scoring stages.

3. Stage Two — Scoring the Vital Signs

Once the intake record is complete, the system evaluates each vital sign individually against established clinical thresholds and assigns a severity score. This approach is inspired by the National Early Warning Score 2 framework, in which each parameter contributes points that reflect how far it deviates from the normal range (Royal College of Physicians, 2017). The respiratory rate is given the greatest weight in

this process, because research consistently demonstrates that an abnormal respiratory rate is the earliest and most sensitive indicator of patient deterioration, yet it remains the vital sign most frequently underdocumented in clinical practice (Cretikos et al., 2008). A respiratory rate that falls at or below eight breaths per minute or rises to twenty-five or above attracts the maximum score of three; a rate between nine and eleven or between twenty-one and twenty-four scores two; and a rate between twelve and twenty, which is the normal adult range, scores zero. The oxygen saturation follows a similar logic, with a saturation at or below ninety-one percent scoring three and normal values above ninety-six scoring zero. The heart rate, systolic blood pressure, and temperature are each scored on the same zero-to-three scale using their respective normal and abnormal thresholds. The Glasgow Coma Scale is scored heavily at the critical end, with a GCS of eight or below or an AVPU of Pain or Unresponsive attracting the maximum three points, reflecting the urgent neurological danger those states represent. Finally, severe pain, defined as a score of eight or above on the visual analogue scale, contributes two additional points. The individual scores for all parameters are added together to produce a single composite vital sign score. For patients under sixteen years of age, the system automatically applies age-banded reference ranges sourced from Fleming et al. (2011), because children's normal vital sign values differ substantially from those of adults and using adult thresholds would produce systematic errors in paediatric triage.

4. Stage Three — Classifying the Chief Complaint

In parallel with vital sign scoring, the free-text chief complaint entered at registration is processed by a natural language classifier trained on the CEDIS Presenting Complaint List. The classifier maps the narrative entry to a standardised complaint code, which the system then checks against a tiered risk library. This step is essential because some of the most dangerous conditions in emergency medicine, including acute coronary syndrome, stroke, anaphylaxis, and major trauma, can present in patients whose vital signs are still within normal limits at the moment of triage. Relying on vital signs alone would risk under-triaging these patients until they deteriorate visibly. The complaint library therefore assigns each complaint category not only a numerical score contribution but also a mandatory minimum triage level. Complaints in the highest risk group, such as cardiac arrest, airway obstruction, and major trauma, enforce a minimum level of one. High-risk presentations including chest pain of suspected cardiac origin, stroke, severe breathlessness, and anaphylaxis enforce a minimum of level two. Moderately urgent complaints such as abdominal pain, suspected fractures, febrile children under five, and acute psychiatric presentations enforce a minimum of level three. Minor complaints such as small lacerations, dental pain, and cold symptoms carry no enforced minimum beyond level four or five. Any complaint code that the system does not recognise is conservatively defaulted to a minimum of level three, to ensure that unclassified presentations are not inadvertently assigned low priority.

5. Stage Four — Calculating the Final Triage Level

With both the vital sign score and the complaint classification in hand, the system calculates a composite acuity score. The vital sign component carries seventy percent of the weight and the complaint component carries thirty percent, reflecting the primary diagnostic value of objective physiological data while still accounting for the clinical significance of the presenting problem. The weighted composite score is then mapped to a provisional CTAS level: the highest scores correspond to level one, and progressively lower scores correspond to levels two through five. However, this

provisional level is immediately checked against the minimum level mandated by the chief complaint classification. If the complaint demands a higher acuity than the vital sign score alone would suggest, the complaint minimum takes precedence, and the patient is assigned the higher acuity level. This floor mechanism is the system's primary safeguard against under-triage. Two further age-based modifiers are then applied: infants under two years of age are automatically escalated to at least level two, and patients aged seventy-five and over are escalated to at least level three, recognising that both groups are at elevated risk of rapid deterioration and may not display the same physiological warning signs as younger adults.

6. Stage Five — Output, Escalation, and Reassessment

The final triage level is displayed on the nurse workstation with colour coding consistent with international emergency triage standards: red for level one, orange for level two, yellow for level three, green for level four, and blue for level five. The display also shows the individual clinical flags that were raised during scoring, such as critical hypoxia, severe hypotension, or neurological compromise, giving the nurse immediate visibility of the specific concerns driving the recommendation. The system then initiates the appropriate escalation pathway automatically. A level one patient triggers an immediate overhead team alert and assigns a resuscitation bay, with a physician notified on a stat basis. A level two patient is directed to the acute care zone with a fifteen-minute wait timer activated; if the patient remains unseen when the timer expires, a second alert is generated. Levels three through five trigger proportionally less urgent bed assignments and wait timers of thirty, sixty, and one hundred and twenty minutes respectively. Level five patients additionally receive a prompt suggesting that their complaint may be appropriate for management in a community pharmacy or general practice setting. All triage decisions, timestamps, vital signs, flags, and assigned personnel are written to the electronic patient record and to a separate audit log at the moment of output. Before any record is committed, however, the nurse must actively confirm the triage level on screen. If the nurse disagrees with the system recommendation, they can select a different level and are prompted to enter a brief reason; this override is recorded in full in the audit log, maintaining transparency and supporting clinical governance review. Finally, the system schedules automatic reassessment prompts for all waiting patients at intervals calibrated to their triage level, ensuring that any patient who deteriorates while waiting is identified before their maximum safe wait time is exceeded.

7. Design Considerations

Several important limitations and design principles underpin this system. First and foremost, the algorithm is advisory only. No triage decision is recorded without explicit nurse confirmation, and the override mechanism is intentional rather than incidental; it reflects the understanding that algorithms cannot account for all the contextual and observational information a skilled nurse brings to a triage encounter. Second, the system must be periodically audited for differential misclassification rates across patient demographic groups, as pain scoring in particular is subject to cultural variation and implicit bias. Third, because respiratory rate is the most frequently mis-documented vital sign in emergency practice, the system includes a prompt encouraging nurses to perform a manual sixty-second count when an automated measurement is unavailable or implausible, rather than accepting an estimated value. Fourth, the natural language classifier used for complaint coding will require ongoing retraining as new presenting complaint patterns emerge. Any complaint it cannot confidently classify is routed to a

conservative default rather than a low-acuity assumption, prioritising patient safety over operational efficiency.

References

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